

1 General Setup

Lemma 1.1 (*Cursed Inequality Lemma*) Let $k \in \mathbb{R}^V$ be a discrete curvature of a Euclidean or Hyperbolic polyhedral metric on a triangulated surface $(S, \mathcal{T})$. Then for all $I \in 2^V - \{\emptyset, V\}$

$$\sum_{v \in I} k(v) > \pi \left(2|I| - \sum_{i=1}^3 |F_i(I)| \right) \quad (1)$$

holds where

$$F_i(I) = \{\text{set of all faces with } i \text{ vertices in } I\} \quad (2)$$

Proof Note $F_i(I)$ and $|F_i(I)|$ may be used interchangeably depending on context.

$$\begin{aligned} \sum_{v \in I} k(v) &= 2\pi|I| - \sum_{\triangle v_1 v_2 v_3 \in F_3(I)} a_1 + a_2 + a_3 \\ &\quad - \sum_{\triangle v_1 v_2 v_3 \in F_2(I)} a_1 + a_2 - \sum_{\triangle v_1 v_2 v_3 \in F_1(I)} a_1 \\ &= 2\pi|I| - \pi F_3(I) - \sum_{\triangle v_1 v_2 v_3 \in F_2(I)} a_1 + a_2 - \sum_{\triangle v_1 v_2 v_3 \in F_1(I)} a_1 \\ &> 2\pi|I| - \pi F_3(I) - \pi F_2(I) - \pi F_1(I) \\ &= \pi \left(2|I| - \sum_{i=1}^3 |F_i(I)| \right) \end{aligned} \quad (3)$$

□

2 Köbe, Andreiev, and Thurston Theorem

Lemma 2.1 Let $(S^2, \mathcal{T})$ be a simplicially-triangulated. Let $v_1, v_2, v_3 \in V = V(\mathcal{T})$ where $\triangle v_1 v_2 v_3 \in \mathcal{T}$ Let $k : V \rightarrow \mathbb{R}$ where

$$k(v) := \begin{cases} \frac{4\pi}{3} & \text{if } v \in [v_1, \dots, v_n] \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

k is in the space of discrete curvatures of associated with the triangulation.

Proof

Let $g : 2^V \rightarrow \mathbb{R}$ where $\forall A \subseteq V$,

$$g(A) = \sum_{v \in A} k(v) \quad (5)$$

We need to show 2 things:

1. Gauss Bonnet:

$$g(V) = g(\{v_1, v_2, v_3\}) = 2\chi(S)\pi \quad (6)$$

2. For all $I \subseteq V$, $g(I) > \pi(2|I| + \sum_{i=1}^3 F_i(I))$:

1. If $|I| < |V| - 2$: $g(I) > 0 = \pi(2|I| + \sum_{i=1}^3 F_i(I))$ by the contrapositive of 6.14 (Notes by professor).
2. If $|I| = |V| - 2$: By Theorem 6.14
 1. $2|I| + \sum_{i=1}^3 F_i(I) = 0$: $|I \cap \{v_1, v_2, v_3\}| \geq n - 2$. Thus $g(I) > 0 = 2|I| + \sum_{i=1}^3 F_i(I)$.
 2. $2|I| + \sum_{i=1}^3 F_i(I) = 2$: Thus $I = V - \{u, v\}$ where u and v have no edge between them. Thus $v_1, v_2, v_3 \in I \Rightarrow \text{true}$
3. If $|I| = |V| - 1$: $|I \cap \{v_1, \dots, v_n\}| \geq n - 1$. Thus $g(I) \geq \frac{8\pi}{3} > 2\pi$ by 6.14 (Notes by professor).

Thus k is a discrete curvature of $(S, \mathcal{T})$. □

Lemma 2.2 Given three pairwise tangent round disks D_1, D_2, D_3 on the Riemann Sphere, there exists a Mobius transformation that sends $D_1 \rightarrow \{z \in \mathbb{C} \mid \Re(z) \leq -1\}$, $D_2 \rightarrow \mathbb{D}$, and $D_3 \rightarrow \{z \in \mathbb{C} \mid \Re(z) \geq 1\}$

Proof Let C be the circle that intersects the 3 tangent points. Let M be the Mobius transformation which sends $C \rightarrow \mathbb{R}$ (x axis). M is the mobius transformation that satisfies the lemma. □

Theorem 2.3 (Köbe-Andreev-Thurston) Let $(S, \mathcal{T})$ be a simplicially-triangulation of 2-sphere. Then there exists a circle packing $\mathcal{P}$ on S^2 , unique up to Mobius transformations, whose nerve is isomorphic to the 1-skeleton of $\mathcal{T}$.

Proof Remove a triangle $\triangle v_1 v_2 v_3$ from $\mathcal{T}$ to obtain a simplicial triangulation $\mathcal{T}_1$ of the topological triangle $\triangle = S^2 - \text{int}(\tau)$. The triangulation has 3 boundary vertices v_1, v_2, v_3 . Consider the double over the boundary, $(\triangle \cup_{\text{id}|_\partial} \triangle, \mathcal{T}_2)$ which is a simplicial triangulation of S^2 with respect to the involution σ of $\mathcal{T}_2$ where each $\mathcal{T}_1$ is interchanged and v_i is fixed. Let $\hat{k} : V \rightarrow \mathbb{R}$ where $\hat{k}(v) = 0$ if $v \neq v_i$ and otherwise $\hat{k}(v) = \frac{4\pi}{3}$. By Lemma 1.1, $\hat{k}$ is a valid discrete curvature of $(S^2, \mathcal{T}_2)$. Let $r : V(\mathcal{T}_2) \rightarrow \mathbb{R}$ be the circle packing metric which induces $\hat{k}$. Since $\mathcal{T}_2$ is symmetric about σ and since $\hat{k} \circ \sigma = \hat{k} \Rightarrow r = r \circ \sigma$ by uniqueness of circle packing metric. Since $\hat{k}(v_1) = \hat{k}(v_2) = \hat{k}(v_3) \Rightarrow \triangle v_1 v_2 v_3$ is equilateral. Furthermore, $r|_{V(\mathcal{T}_1)}$ is a circle packing metric on $(\triangle, \mathcal{T}_1)$ with zero discrete curvature on all interior vertices and $r(v_1) = r(v_2) = r(v_3)$. Thus the circle packing on the complex plane whose nerve is $\mathcal{T}_1$, which is the same as the 1 skeleton of $\mathcal{T}$.

To show the uniqueness of the circle packing, suppose P and Q are circle packings on S^2 (Riemann Sphere) whose nerves are isomorphic to the 1-skeleton of $\mathcal{T}$. Take a triangle $\triangle v_1 v_2 v_3$ in $\mathcal{T}$ and denote the corresponding tangent disks in P and Q by $\{A_1, A_2, A_3\}$ and $\{B_1, B_2, B_3\}$, respectively. By Lemma 1.2, there is a M , mobius transformation, which sends $A_i \rightarrow B_i$. We may assume A_i and B_i are disks of radius 1 such that ∞ is in the unbounded component of $\mathbb{C} \cup \{\infty\} - \bigcup_{j=1}^3 A_j$. Then $P - \{A_1, A_2, A_3\}$ and $Q - \{B_1, B_2, B_3\}$ comes from the circle packing metric on triangulated $(\triangle, \mathcal{T})$ with same discrete curvature and radii at boundary vertices are 1. By Rigidity Theorem for Euclidean background, the circle packing metrics are the same or $P = Q$ □

3 Hyperbolic Theorems

Theorem 3.1 (Hyperbolic Gauss Bonnet) Let $(S, \mathcal{T}, l)$ be a compact triangulated hyperbolic polyhedral surface with or without boundary. Then its area is equal $\sum_{v \in V} k(v) - 2\pi\chi(S)$.

Proof Without loss of generality, S is closed by doubling it along its boundary ie consider $S' = S \cup_{\text{id}|_{\partial}} S$. Suppose τ is a triangle and v is a vertex. Then if α is a angle in the triangulation. $\alpha > \tau$ means α is a angle in the triangle and $\alpha > v$ α is angle of v .

$$\begin{aligned}
\sum_{v \in V} k(v) - 2\pi\chi(S) &= 2\pi|V| - \sum_{\text{all angles } \alpha} \alpha - 2\pi(|V| - |E| + |F|) \\
&= 2\pi|E| - 2\pi|F| - \sum_{\text{all angles } \alpha} \alpha \quad (2|E| = 3|F|) \\
&= \pi|F| - \sum_{\tau \in \mathcal{T}} \sum_{\alpha > \tau} \alpha \\
&= \sum_{\tau \in \mathcal{T}} \left(\pi - \sum_{\alpha > \tau} \alpha \right) \\
&= \sum_{\tau \in \mathcal{T}} \text{Area}(\tau) \\
&= \text{Area}(\mathcal{T})
\end{aligned} \tag{7}$$

□

Lemma 3.2 (*Colin De Verdiere's Variational Principle*) Let a hyperbolic triangle have edge lengths $l_i = r_j + r_k$, $\{i, j, k\} = \{1, 2, 3\}$, and angles a_1, a_2, a_3 such that a_i is opposite to l_i . Suppose $r_i > 0$ and $u_i = \int_{r_i}^{\infty} \frac{dt}{\sinh(t)}$ then

1. $\frac{\partial a_i}{\partial u_j} = \frac{\partial a_j}{\partial u_i} < 0$ and $\frac{\partial a_i}{\partial u_i} > 0$
2. $\frac{\partial(a_1 + a_2 + a_3)}{\partial u_i} > 0$
3. The function $W_{cp}(u_1, u_2, u_3) = \int_0^u \sum_{j=1}^3 a_j du_j$ is a strictly concave function of $u = (u_1, u_2, u_3)$ in $R_{>0}^3$ satisfying $\frac{\partial W_{cp}}{\partial u_i} = a_i$
4. Area of a hyperbolic triangle is strictly increasing function

Theorem 3.3 (*Hyperbolic Thurston's Uniqueness Theorem*) Two hyperbolic circle packing metrics on a compact triangulated surface $(S, \mathcal{T})$ with the same discrete curvatures are the same

Proof Without loss of generality assume that S is closed, double S across boundary otherwise. Let r and R be circle packing metrics such that $r \neq R$. We want to show that $k_r \neq k_R$. Let $V = \{v_1, \dots, v_n\}$. $\forall f : V \rightarrow \mathbb{R}$, $f_i := f(v_i)$. Let $u, u' \in \mathbb{R}^V$ such that $u_i = \int_{r_i}^{\infty} \frac{dt}{\sinh(t)}$ and $u'_i = \int_{R_i}^{\infty} \frac{dt}{\sinh(t)}$. Thus $a - b \neq 0$. Define $W : \mathbb{R}^V \rightarrow \mathbb{R}$ by

$$W(x) = \sum_{\triangle v_i v_j v_k \in F} W_{cp}(x_i, x_j, x_k) \tag{8}$$

Since W_{cp} is strictly concave, W is strictly concave. Furthermore since $\frac{\partial W_{cp}}{\partial x_i} = a_i$ where a_i is the angle in $\triangle v_i v_j v_k$, we have

$$\frac{\partial W}{\partial x_i}(x) = \sum_{a > v_i} a = 2\pi - k_{f(x)}(v_i) \tag{9}$$

where f is the inverse function of $\int_r^{\infty} (\sinh(t))^{-1} dt$.

Thus $(\nabla W)(x) = 2\pi(1, \dots, 1) - k_{f(x)}$

Take $g(t) = W(tu + (1-t)u')$. g is strictly concave because W is strictly concave thus $g'(0) \neq g'(1)$. But $g'(t) = (\nabla W)(tu + (1-t)u') \cdot (u - u') \Rightarrow \nabla W(u) \neq \nabla W(u') \Rightarrow k_r \neq k_R$. □

Lemma 3.4 For any $\varepsilon > 0$, $\exists N > 0$ such that if $r_1 > N$, $r_2 > 0$, and $r_3 > 0$ then for any hyperbolic edge lengths $l_i = r_j + r_k$, we have a inner angle $a_1 < \varepsilon$

Proof Let $\triangle v_1 v_2 v_3$ be a hyperbolic triangle. Place $\triangle v_1 v_2 v_3$ in Poincare disk model where v_1 is the origin. Since diameters are geodesics in this model, that means $v_1 v_2$ and $v_1 v_3$ are Euclidean lines. By the construction v_1, v_2, v_3 are the centers of pairwise tangent hyperbolic balls of radius r_1, r_2, r_3 respectively. If r_1 is large then the euclidean radius of B_1 is large which means the euclidean radius of B_2, B_3 is small. Then $a_1 \rightarrow 0$. So as $r_1 \rightarrow \infty \Rightarrow a_1 \rightarrow 0 \Rightarrow$ lemma follows from standard limit argument. $\square$

Theorem 3.5 (*Characterization of Discrete Curvature for Hyperbolic Circle packing Metrics*)

Let $(S, \mathcal{T})$ be a closed triangulated surface. Then the set of all discrete curvatures of a hyperbolic circle packing metrics on $(S, \mathcal{T})$ is given by

$$\mathcal{K}_h = \{k \in (-\infty, 2\pi)^V \mid I \subset V \ni I \neq \emptyset, \text{Cursed Inequality holds}\} \quad (10)$$

Proof By the Cursed Inequality Lemma, $K = \mathcal{K}_h$ contains the discrete curvature induced by all hyperbolic circle packing metrics on $(S, \mathcal{T})$. Let $Q = R_{>0}^V$. Let $f : Q \rightarrow K$ be the function that takes a circle packing metric to the discrete curvature it induces. By Thurston's Hyperbolic Discrete Curvature Uniqueness Theorem, f is injective. By the hyperbolic cosine law, f is continous. Since the $Q, K \subset \mathbb{R}^V$ and are connected manifolds of dimension $|V|$, by the invariance of domain theorem, it suffices to prove that $f(Q)$ is a closed subset of K . To do this we want to show that as we approach the boundary of Q when we apply f , we approach the boundary of K . Let $s \in \partial Q \Rightarrow \exists v \in V \ni s(v) = 0$ or $s(v) = \infty$. Define

$$I := \{v \in V \mid s(v) = 0\} \quad (11)$$

If $I = \emptyset \Rightarrow s(v) = \infty$. As $r \rightarrow s$, $k_r(v) \rightarrow 2\pi$. Thus $f(r) \rightarrow \partial K$.

If $I \neq \emptyset$, as $r \rightarrow s$

$$\begin{aligned} \sum_{v \in I} k_r(v) &= 2\pi|I| - \sum_{v \in I} \sum_{a > v} a \\ &= \pi \left(2|I| - \sum_i F_i(I) \right) \end{aligned} \quad (12)$$

thus $f(r) \rightarrow \partial K$ $\square$

4 Discrete Schwartz-Alfhors-Pick Lemma

Theorem 4.1 (*Discrete Shwartz-Alfhors-Pick Lemma*) Let $(S, \mathcal{T})$ be a closed connected simplicially-triangulated surface such that its vertex set $V(\mathcal{T}) = V_1 \sqcup V_2$ such that $V_1 \neq \emptyset$ and in the case Euclidean background, $V_2 \neq \emptyset$. Suppose R and r are two scircle packing radius assignments on $(S, \mathcal{T})$ with either Euclidean or hyperbolic background such that $R|_{V_2} \geq r|_{V_2}$ and their curvatures k_R and k_r satisfy $k_R|_{V_1} \geq k_r|_{V_1}$. Then $R \geq r$ and for any two vertices v_1 and v_2

$$d_R(v_1, v_2) \geq d_r(v_1, v_2) \quad (13)$$

Proof Let

$$X := \{x \in \mathbb{R}_{>0}^V \mid x|_{V_2} \geq r|_{V_2} \text{ and } k_x|_{V_1} \geq k_r|_{V_1}\} \quad (14)$$

Note $r \in X$ and by definition of R , $R \in X$. We want to prove that $\forall v \in V$, $r(v) = \inf\{x(v) \mid x \in X\}$ i.e $r = \inf\{x \mid x \in X\}$. Thus by definition $R \geq r$.

1. $x, y \in X \Rightarrow \min\{x, y\} \in X$: $\forall x, y \in X$, let $z = \min\{x, y\} = (\min(x_v, y_v))_{v \in V}$. $\forall v \in V_2$, $z(v) = x(v) \geq r(v)$ or $z(v) = y(v) \geq r(v)$. Thus $z(v) \geq r(v)$. Thus $z|_{V_2} \geq r|_{V_2}$. $\forall v \in V_1$, without loss of generality, $w(v) = x(v)$. $\forall \triangle va_1a_2 \in \mathcal{T}$, $w(a_1) \leq x(a_1)$ and $w(a_2) \leq x(a_2)$. By the Variational Principle for Euclidean and Hyperbolic Triangles, $\frac{\partial \theta(v)}{\partial x(a_i)} > 0 \Rightarrow \frac{\partial k(v)}{\partial r(a_i)} < 0$. In other words, not increasing the radii of all the vertices which share an edge with v will force the discrete curvature at v to increase. Since $x(a_i) \geq w(a_i)$ for all such triangles while $w(v) = x(v) \Rightarrow k_w(v) \geq k_x(v) \geq k_r(v)$. Thus $z \in X$.
2. Now suppose $w = \inf X$. For each $v \in V$, let $(x_{n,v})_{n \in \mathbb{N}}$ be a sequence in X such that $w(v) = \lim_n x_n(v)$. Let $V = \{v_1, \dots, v_m\}$ and define $x_n = \min\{x_{n,v_i} \mid i \in [1, \dots, m]\}$. By 1. $x_n \in X$. Since $x_{n,v_i} \geq x_n$,

$$\limsup_{n \rightarrow \infty} x_n(v) \leq \limsup_{n \rightarrow \infty} x_{n,v}(v) = \lim_{n \rightarrow \infty} x_{n,v}(v) = w(v) \quad (15)$$

On the other hand, $w(v) \leq x_n(v)$. Thus

$$w(v) = \liminf_{n \rightarrow \infty} x_n(v) \quad (16)$$

. Combining both inequalities, we see

$$w = \lim_{n \rightarrow \infty} x_n \quad (17)$$

Now we have to show that $w \in X$.

1. $w|_{V_2} \geq r|_{V_2}$: Assume the contrary $\exists v \in V_2 \ni w(v) < r(v)$. Thus $\exists x \in X \ni w(v) \leq x(v) < r(v)$ by definition of infimum. But this is a contradiction, since $x \in X$ and $x(v) \geq r(v)$. Thus $\forall v \in V_2$, $w(v) \geq r(v)$.
2. $w \in R_{>0}^V$: Assume the contrary that $I = \{v \in V \mid w(v) = 0\} \neq \emptyset$. By 6.11 (6.12) and Characterization of Discrete Curvature for Hyperbolic Circle Packings,

$$\lim_{n \rightarrow \infty} \sum_{v \in I} k_{x_n}(v) = \pi \left(2|I| - \sum_{j=1}^3 |F_j(I)| \right) \quad (18)$$

Assume the contrary, $\exists v \in I \cap V_2$. $\forall n \in \mathbb{N}$, $x_n(v) \geq r(v) > 0 \Rightarrow w(v) = \lim_{n \rightarrow \infty} x_n(v) \geq r(v) > 0$. However this is a contradiction since $w(v) = 0$ since $v \in I$. Thus $I \cap V_2 = \emptyset$.

Thus $\forall v \in I$, $k_{x_n}(v) \geq k_r(v)$. By the Cursed Inequality,

$$\sum_{v \in I} k_{x_n}(v) \geq \sum_{v \in I} k_r(v) > \pi \left(2|I| - \sum_{j=1}^3 |F_j(I)| \right) \quad (19)$$

which means,

$$\lim_{n \rightarrow \infty} \sum_{v \in I} k_{x_n}(v) \geq \sum_{v \in I} k_r(v) > \pi \left(2|I| - \sum_{j=1}^3 |F_j(I)| \right) \quad (20)$$

This is a contradiction. Thus $I = \emptyset$ and $w \in R_{>0}^V$.

3. $k_w|_{V_1} \geq k_r|_{V_1}$: $k_w(v) = \lim_n k_{x_n}(v) \geq k_r(v)$. The first equality comes from sequential definition of continuity.

Thus $w \in X$. We now want to show that $w = r$. By 1 and definition of $\inf$, $w|_{V_2} = r|_{V_2}$. We now want to show that $k_w|_{V_1} \leq k_r|_{V_1}$. Assume the contrary, $\exists v \in V_1 \ni k_w(v) > k_r(v)$. For a small $t \in$

$(0, w(v))$, define $w^t \in \mathbb{R}_{>0}^V$ as follows. If $u \in V - \{v\}$, $w^t(u) = w(u)$ and $w^t(v) = w(v) - t > 0$. Since, $v \in V_2 \Rightarrow v \notin V_1 \Rightarrow w^t|_{V_1} \geq r|_{V_1}$. For t small, by continuity we may assume that $k_{w^t}|_{V_1} \geq k_r|_{V_1}$. Thus $w^t \in X$. But $w^t(v) < w(v)$ this contradicts the fact that $w = \inf X$. Thus $k_w|_{V_1} = k_r|_{V_1}$.

We now need to show that $w = r$ using the fact that $w|_{V_2} = r|_{V_2}$ and $k_w|_{V_1} = k_r|_{V_1}$. Let $A = \mathbb{R}$ if the background geometry is Euclidean and $A = (0, \infty)$ if the background geometry is Hyperbolic. Let $\varphi : (0, \infty) \rightarrow A$ where

$$\varphi(x) := \begin{cases} \ln(x) & \text{if Euclidean Background} \\ \int_x^\infty \frac{dt}{\sinh(t)} & \text{if Hyperbolic Background} \end{cases} \quad (21)$$

φ is a smooth bijection so φ^{-1} exists. Let $\mathcal{R} = \{x \in \mathbb{R}_{>0}^V \mid \varphi^{-1}(x)|_{V_2} = r|_{V_2}\}$.

You can view $\mathcal{R} = (0, \infty)^{V_1}$. For any triangle $\triangle v_1 v_2 v_3$, with angles θ_i be the angle of v_i . Let $W_{cp}(x_1, x_2, x_3) = \int_0^x \sum_{i=1}^3 \theta_i(u) du_i$. Let $W : A^V \rightarrow \mathbb{R}$ where

$$W(x) := \sum_{\triangle v_i v_j v_k \in \mathcal{T}} W_{cp}(x_i, x_j, x_k) \quad (22)$$

Let $F = W|_{\mathcal{R}}$. Thus F is a function from $(0, \infty)^{V_1} \rightarrow \mathbb{R}$. Thus

$$\nabla F(x) = 2\pi(1, \dots, 1) - k_{\varphi^{-1}(x)}|_{V_1} \quad (23)$$

$\forall x, y \in \mathcal{R} \subseteq \mathbb{R}^V \ni x \neq y, x(w) = y(w)$. Take $A = \{v \in V \mid x(v) = y(v)\}$ and $B = \{v \in V \mid x(v) \neq y(v)\}$. $A \cap B = \emptyset$. $V_2 \subseteq A$ and since $x \neq y$, $B \neq \emptyset$. Since $(S, \mathcal{T})$ is connected, then the graph formed by the triangulation is connected, which means $\exists(a, b) \in A \times B \Rightarrow \triangle abc \in \mathcal{T}$. $(x(a) - y(a), x(b) - y(b), x(c) - y(c)) = (0, x(b) - y(b), x(c) - y(c)) \neq c(1, 1, 1)$. Thus W_{cp} is strictly concave on $\{t(x(a), x(b), x(c)) - (1-t)(y(a), y(b), y(c))\}$. Thus F is strictly concave on this set.

Thus F is strictly concave. Thus $\nabla F : \mathcal{R} \rightarrow \mathbb{R}^{V_1}$ is injective thus $k_{\varphi^{-1}(x)}|_{V_1} = 2\pi(1, \dots, 1) - \nabla F$ is injective. Since $k_w|_{V_1} = k_r|_{V_1} \Rightarrow r|_{V_1} = w|_{V_1} \Rightarrow r = w$. Thus $R \geq r$. $\square$

5 Growing Hyperbolic Circle Packing

Lemma 5.1 Let $F = \{v; v_1, \dots, v_n\}$ be a closed combinatorial flower. Let $R = \{r; r_1, \dots, r_n\}$ be its hyperbolic label (circle packing metric). Suppose $k(v) \leq 0$, then $r \leq -\log(\sin(\frac{\pi}{n}))$

Proof

1. If $k(v) < 0$: r could increase and the hypothesis will still be satisfied
2. If $r_i < \infty$: Increasing r_i would decrease $k(v)$ (b/c $\frac{\partial k(v)}{\partial r_i} < 0$) thus permitting r to be increased.

Thus case where $k(v) = 0$ and $r_i = \infty$ gives the maximum r . In this case, all petals are horocycles. Thus the angle sum at v is 2π , thus each triangle must contribute $\frac{2\pi}{n}$. Each triangle is $\triangle vv_{i-1}v_i$. Let θ_i denote v 's angle in the triangle.

$$\cos(\theta_i) = \frac{\cosh(r + r_{i-1}) \cosh(r + r_i) - \cosh(r_i + r_{i-1})}{\sinh(r + r_{i-1}) \sinh(r + r_i)} \quad (24)$$

For large x , $\cosh(x) \sim \frac{1}{2}e^x \sim \sinh(x)$. Thus:

$$\begin{aligned}
\frac{\cosh(r + r_{i-1}) \cosh(r + r_i) - \cosh(r_i + r_{i-1})}{\sinh(r + r_{i-1}) \sinh(r + r_i)} &\sim \frac{\frac{1}{4}e^{2r+r_i+r_{i-1}} - \frac{1}{2}e^{r_i+r_{i-1}}}{\frac{1}{4}e^{2r+r_i+r_{i-1}}} \\
&= \frac{\frac{1}{2}e^{r_i+r_{i-1}}(\frac{1}{2}e^{2r} - 1)}{\frac{1}{2}e^{r_i+r_{i-1}}(\frac{1}{2}e^{2r})} \\
&= 1 - 2e^{-2r}
\end{aligned} \tag{25}$$

Thus as $r_i \rightarrow \infty$ and $r > 0$.

$$\theta_i = \arccos(1 - 2e^{-2r}) = \frac{2\pi}{n} \tag{26}$$

$$e^{-2r} = 1 - \cos\left(\frac{2\pi}{n}\right) = \sin^2\left(\frac{\pi}{n}\right) \Rightarrow r = -\ln\left(\sin\left(\frac{\pi}{n}\right)\right) \tag{27}$$

□

Theorem 5.2 Let K be a combinatorial closed disk (a finite simply connected simplicial complex with nonempty boundary). Suppose there exists a circle packing P for K in the unit disk $\mathbb{D}$. Then there exists a univalent circle packing P_K for K in $\mathbb{D}$ (where circle interiors are disjoint) such that all boundary circles are horocycles (tangent to $\partial\mathbb{D}$).

Proof We adapt the Discrete Schwartz Lemma method in reverse. We will have a set which is closed under maximums, whose supremum we will prove is the necessary circle packing metric.

Let $\Phi := \{r \in \mathbb{R}_{>0}^V \mid k_r(v) \leq 0 \text{ for all interior } v\}$. We want to show that $R = \sup(\Phi)$ is the circle packing metric we need.

1. **Nonempty:** Since K has nonempty boundary and P is a valid circle packing, interior vertices have angle sums exactly 2π . Thus the radii r from P belong to Φ .
2. **Closed under Maximum:** Suppose $a, b \in \Phi$. We want to show that $c = \max(a, b) \in \Phi$. Suppose v is a interior vertex. Without loss of generality, $c(v) = a(v)$. $\forall \triangle vu_1u_2$, $c(u_i) \leq a(u_i)$ and $c(u_i) \leq b(u_i)$. $\frac{\partial k(v)}{\partial u_i} < 0 \Rightarrow k_c(v) \leq k_a(v) \leq 2\pi$. Thus $c \in \Phi$.
3. **R finite for all internal vertices:** For all interior v , let n be the degree of the vertices (graph theoretic). For each $r \in \Phi$, $k_r(v) \leq 2\pi$. By Lemma 5.1, $r(v) \leq -\ln(\sin(\frac{\pi}{n}))$. Thus, $R(v) \leq -\ln(\sin(\frac{\pi}{n}))$, $n \geq 2$. Thus $R(v) \leq -\ln(\sin(\frac{\pi}{2})) < \infty$.
4. **Solution:**

□